



Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	3d, pressure-based, SST k-omega
Version	24.1.0-10184
Source Revision	5b3f9fb3c8
Build Time	Nov 22 2023 10:25:54 EST
CPU	12th Gen Intel(R) Core(TM) i5-1235U
OS	Windows

Geometry and Mesh

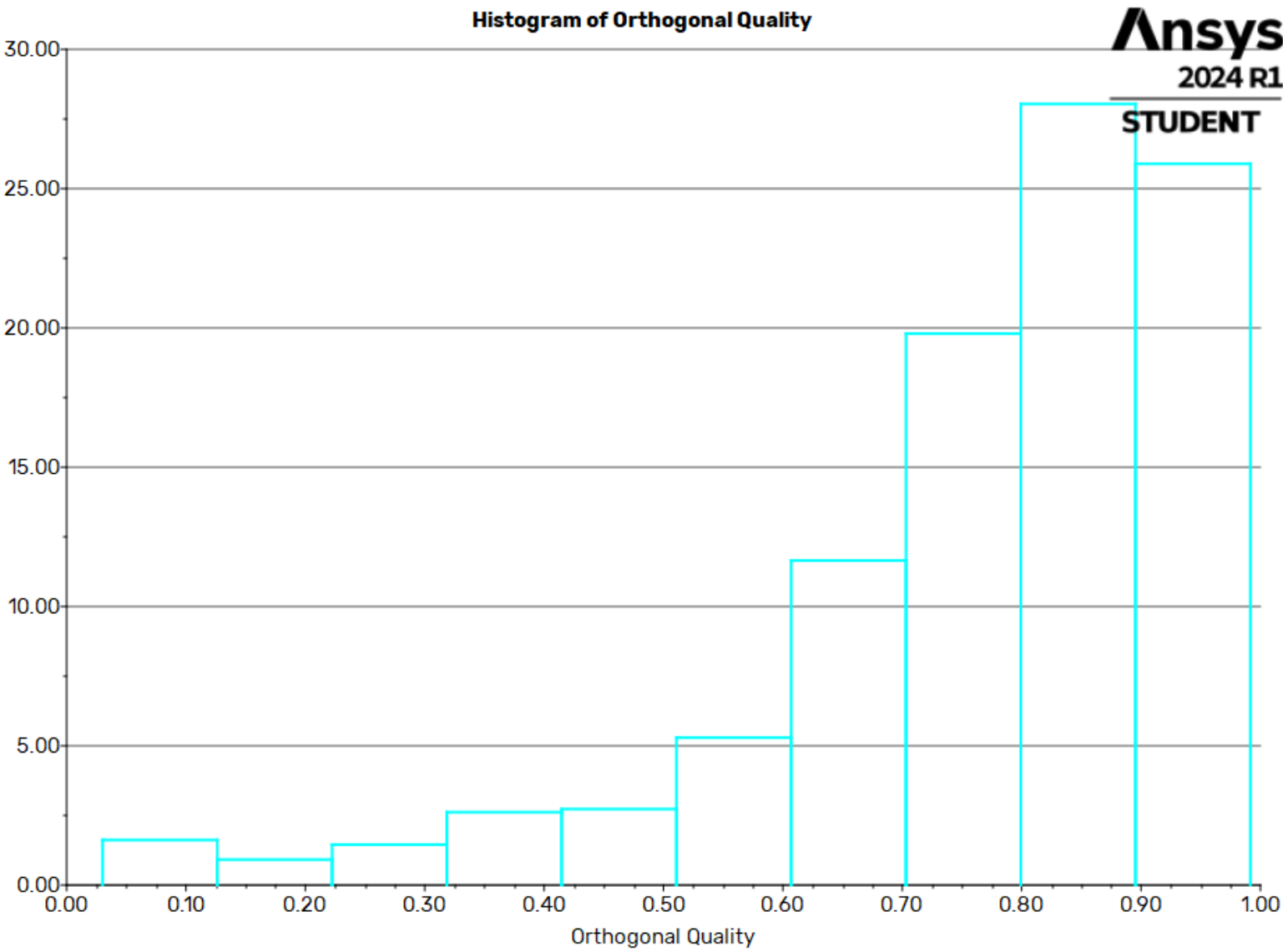
Mesh Size

Cells	Faces	Nodes
502149	1141619	168310

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fff-inner_volume	Mixed Cell	0.22167309	51.380352
fff-outer_volume	Mixed Cell	0.10037538	140.09848
fff-outer_pipe	Mixed Cell	0.036390431	35.760067
fff-inner_pipe	Mixed Cell	0.030077074	79.518288

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	SST k-omega turbulence model
Heat Transfer	Enabled

Material Properties

— Fluid	
— water-liquid	
Density	998.2 kg/m^3
Cp (Specific Heat)	4182 J/(kg K)
Thermal Conductivity	0.6 W/(m K)
Viscosity	0.001003 kg/(m s)
Molecular Weight	18.0152 kg/kmol
— air	
Density	1.225 kg/m^3
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	1.7894e-05 kg/(m s)
Molecular Weight	28.966 kg/kmol
— Solid	
— wood	
Density	700 kg/m^3
Cp (Specific Heat)	2310 J/(kg K)
Thermal Conductivity	0.173 W/(m K)
— aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

— Fluid	
— fff-inner_volume	
Material Name	water liquid
Specify source terms?	no

Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no
— fff-outer_volume	
Material Name	water liquid
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no
— Solid	
— fff-outer_pipe	
Material Name	wood
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Solid Motion?	no
— fff-inner_pipe	
Material Name	aluminum
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Solid Motion?	no

Boundary Conditions

— Inlet	
— inlet_outer_pipe	
Reference Frame	Absolute
Mass Flow Specification Method	Mass Flow Rate
Mass Flow Rate [kg/s]	0.1008
Total Temperature [C]	28
Supersonic/Initial Gauge Pressure [Pa]	0
Direction Specification Method	Normal to Boundary
Turbulent Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
— inlet_inner_pipe	
Reference Frame	Absolute
Mass Flow Specification Method	Mass Flow Rate
Mass Flow Rate [kg/s]	0.1008
Total Temperature [C]	56
Supersonic/Initial Gauge Pressure [Pa]	0

Direction Specification Method	Normal to Boundary
Turbulent Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
— Outlet	
— outlet_inner_pipe	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Total Temperature [C]	26.85
Backflow Direction Specification Method	Normal to Boundary
Turbulent Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
— outlet_outer_pipe	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Total Temperature [C]	26.85
Backflow Direction Specification Method	Normal to Boundary
Turbulent Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
— Wall	
— interior-fff-outer_pipe-fff-outer_volume-shadow	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	wood
Thermal BC Type	Coupled
Enable shell conduction?	no
Convective Augmentation Factor	1
— interior-fff-outer_pipe-fff-outer_volume	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	wood

Thermal BC Type	Coupled
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	rough bc standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
— outer_surface_inner_pipe-fff-inner_pipe-fff-outer_pipe-shadow	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Enable shell conduction?	no
Convective Augmentation Factor	1
— outer_surface_inner_pipe-fff-inner_pipe-fff-outer_pipe	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Enable shell conduction?	no
Convective Augmentation Factor	1
— outer_surface_inner_pipe-fff-inner_pipe-fff-outer_volume-shadow	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Enable shell conduction?	no
Convective Augmentation Factor	1
— outer_surface_inner_pipe-fff-inner_pipe-fff-outer_volume	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	rough bc standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
— interior-fff-inner_pipe-fff-inner_volume-shadow	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum

Thermal BC Type	Coupled
Enable shell conduction?	no
Convective Augmentation Factor	1
— interior-fff-inner_pipe-fff-inner_volume	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Coupled
Enable shell conduction?	no
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	rough bc standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Convective Augmentation Factor	1
— outer_surface_outer_pipe	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	wood
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Enable shell conduction?	no
Convective Augmentation Factor	1
— outer_surface_inner_pipe-fff-inner_pipe	
Wall Thickness [m]	0
Heat Generation Rate [W/m^3]	0
Material Name	aluminum
Thermal BC Type	Heat Flux
Heat Flux [W/m^2]	0
Enable shell conduction?	no
Convective Augmentation Factor	1

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 C
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fff-inner_volume

Solver Settings

— Equations	
Flow	True
Turbulence	True
Energy	True
— Numerics	
Absolute Velocity Formulation	True
— Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Turbulent Kinetic Energy	0.75
Specific Dissipation Rate	0.75
Turbulent Viscosity	1
Energy	0.75
Explicit Momentum	0.5
Explicit Pressure	0.5
— Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
— Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
Energy	Second Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [C]	-272.15
Maximum Static Temperature [C]	4726.85
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	4
Case Read	34.282 seconds
Data Read	4.686 seconds
Virtual Current Memory	1.64302 GB
Virtual Peak Memory	1.64735 GB
Memory Per M Cell	2.03969

Solution Status

Iterations: 200

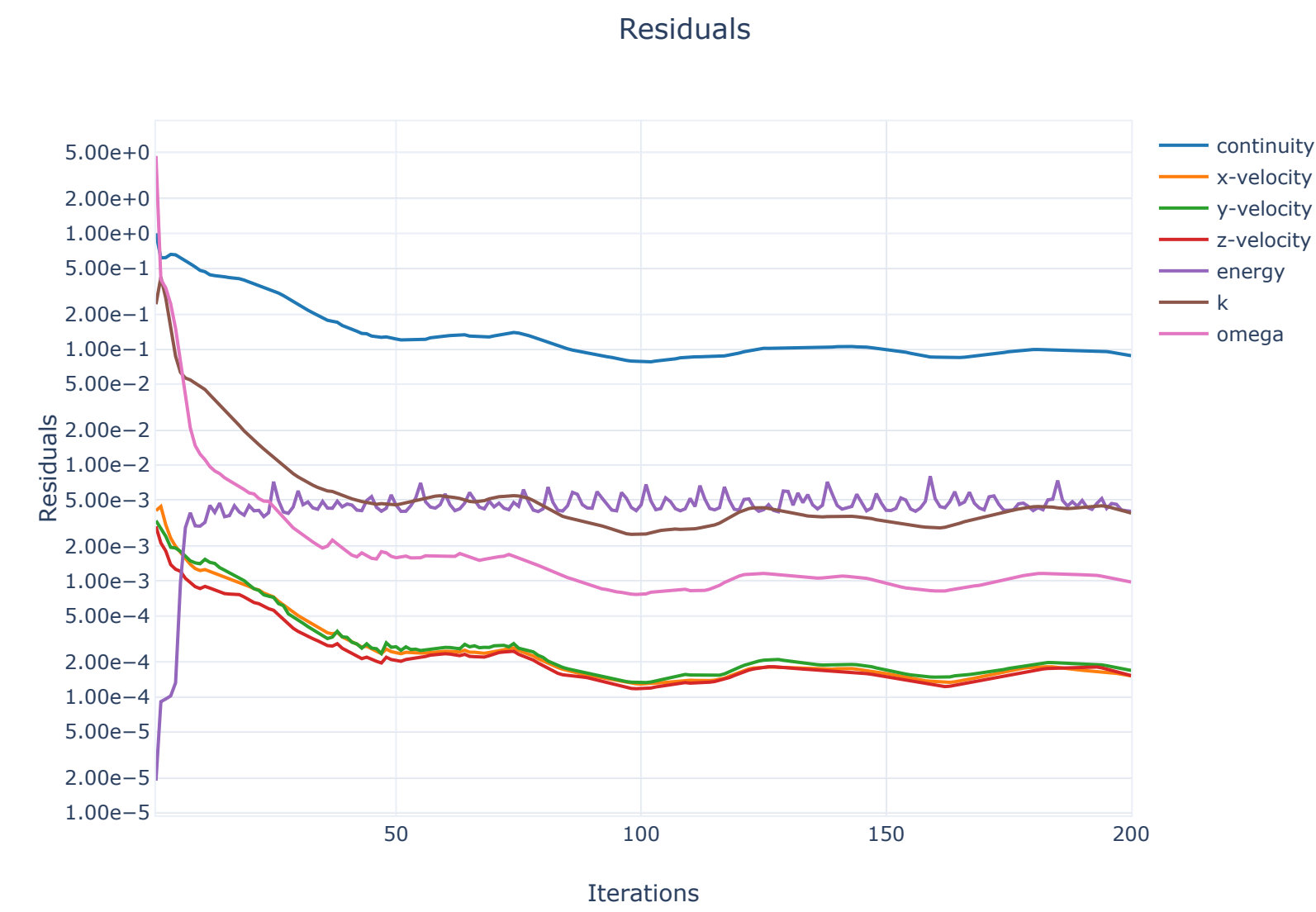
	Value	Absolute Criteria	Convergence Status
continuity	0.08800725	0.001	Not Converged
x-velocity	0.0001517821	0.001	Converged
y-velocity	0.0001692143	0.001	Converged
z-velocity	0.000152761	0.001	Converged
energy	0.003995522	1e-06	Not Converged
k	0.003856884	0.001	Not Converged
omega	0.0009802725	0.001	Converged

Report Definitions

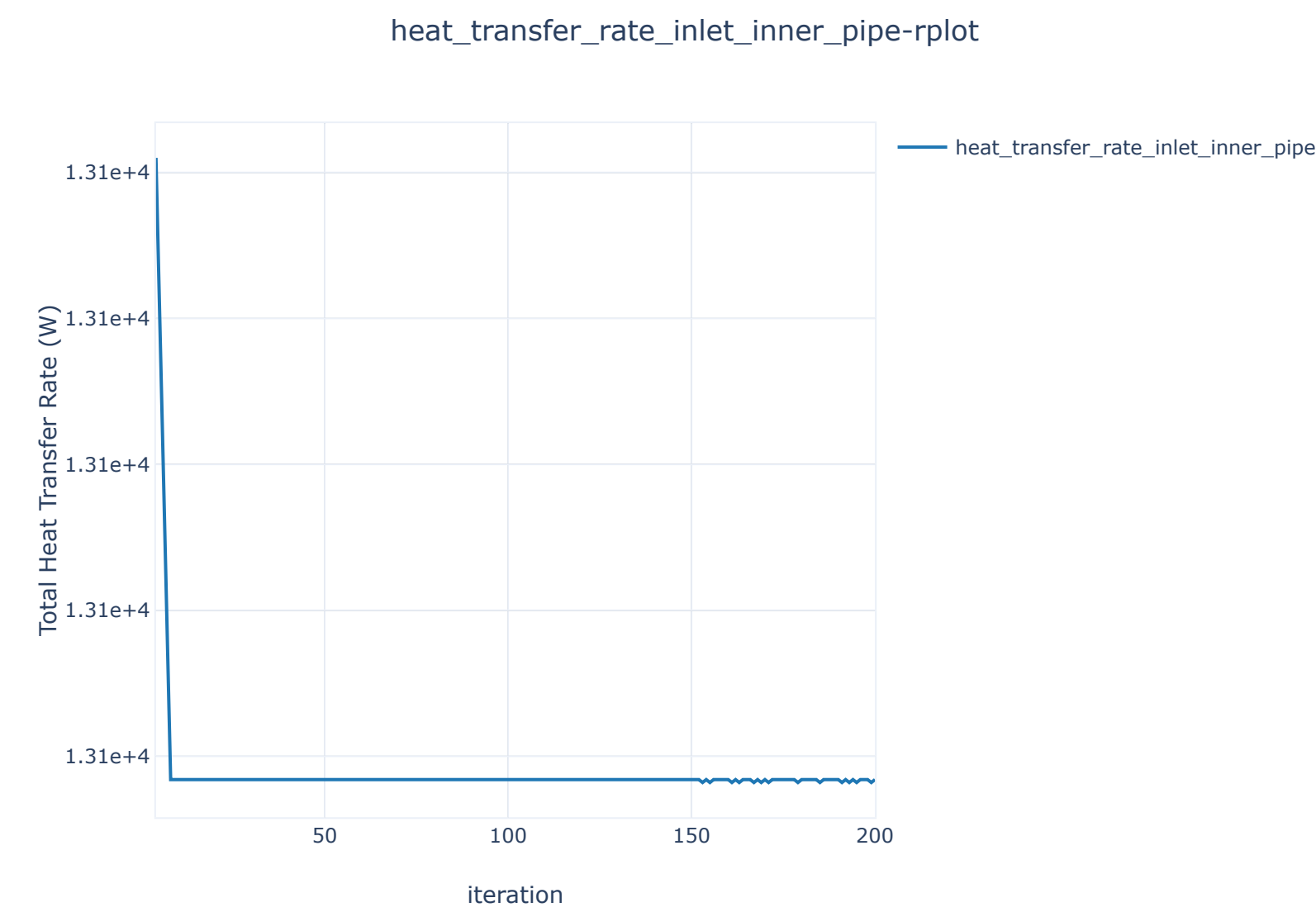
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heat_transfer_rate_outlet_inner_pipe	-12252	W
heat_transfer_rate_inlet_outer_pipe	1264.634	W
heat_transfer_rate_inlet_inner_pipe	13067.92	W

Plots

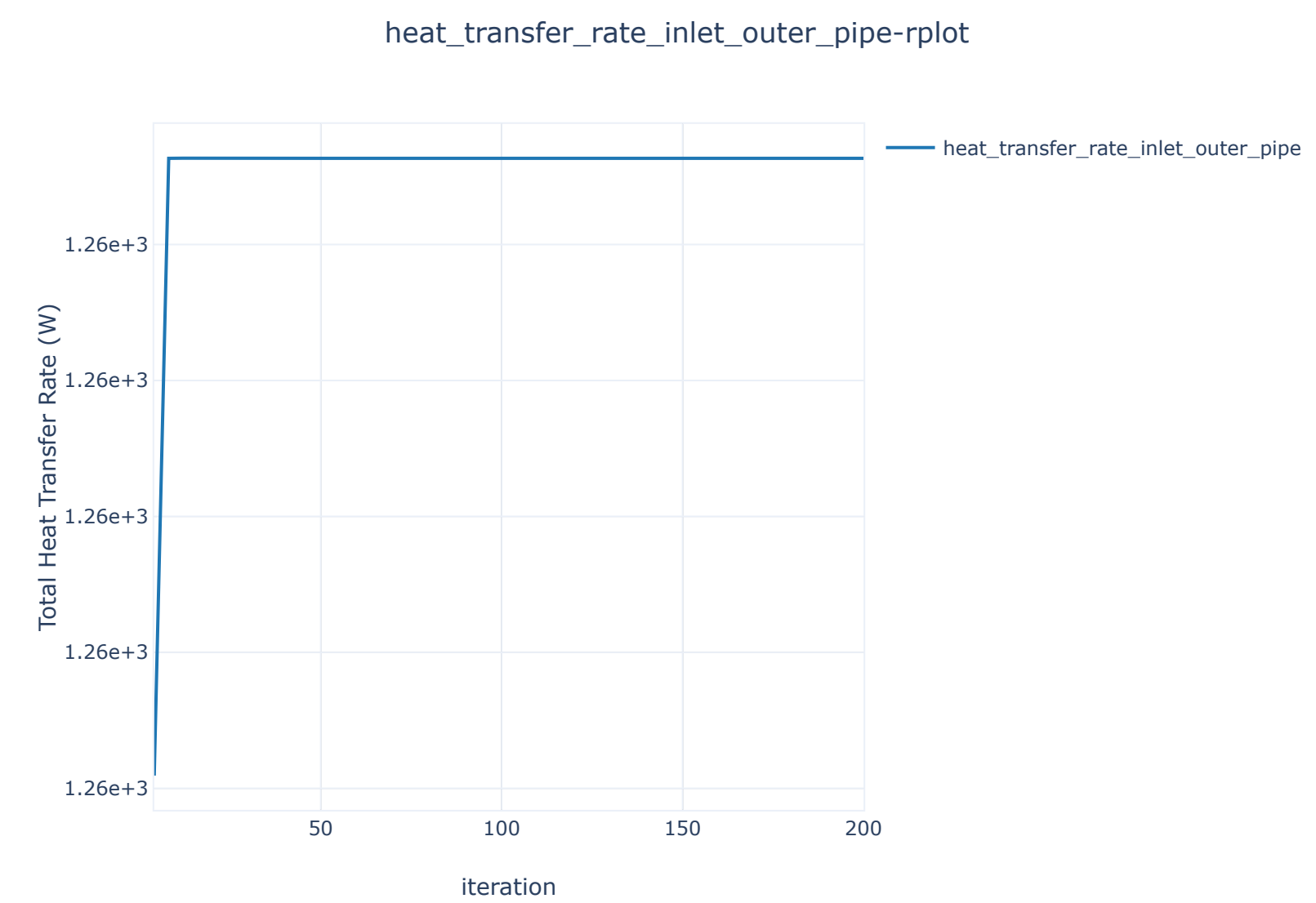
Residuals



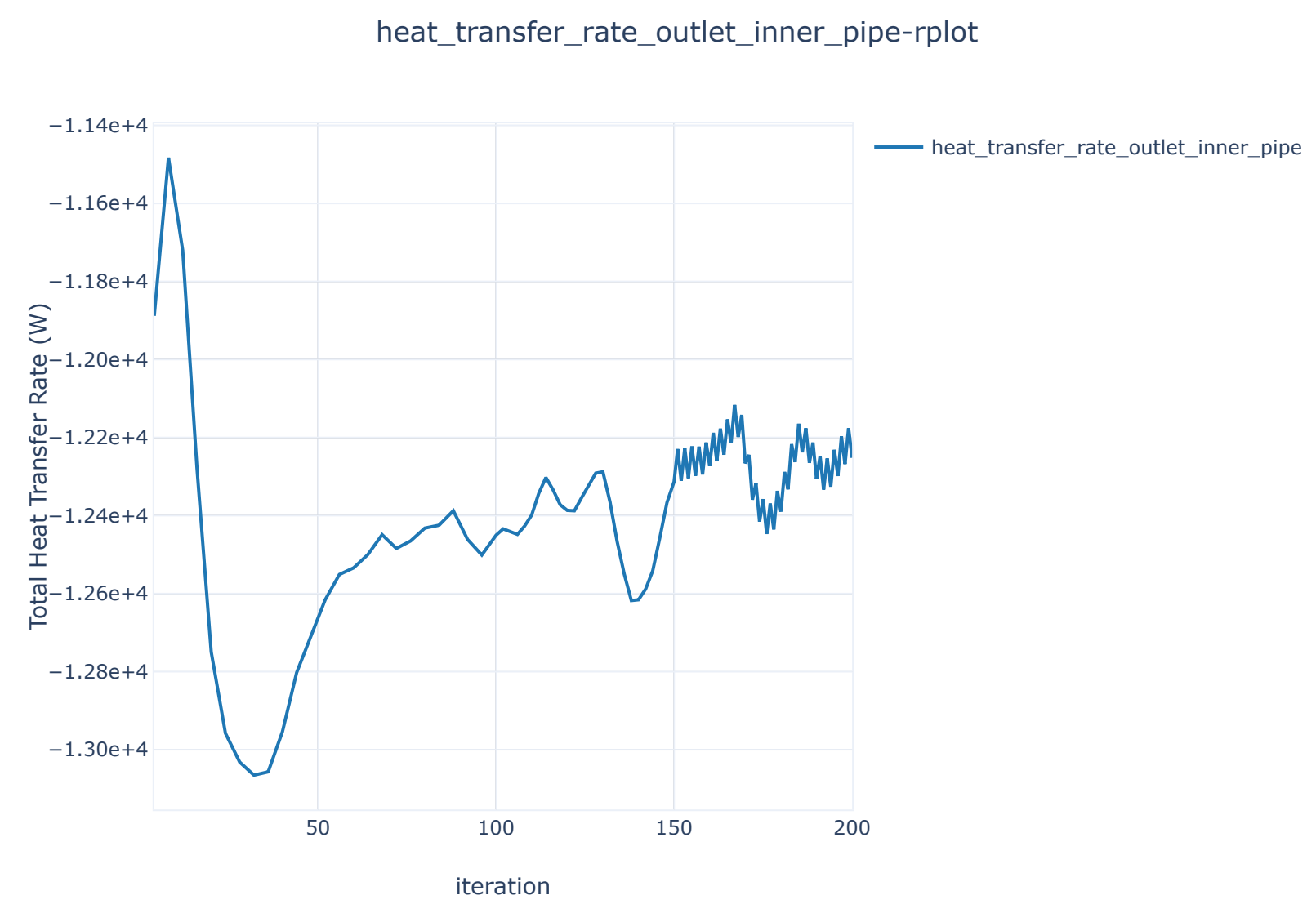
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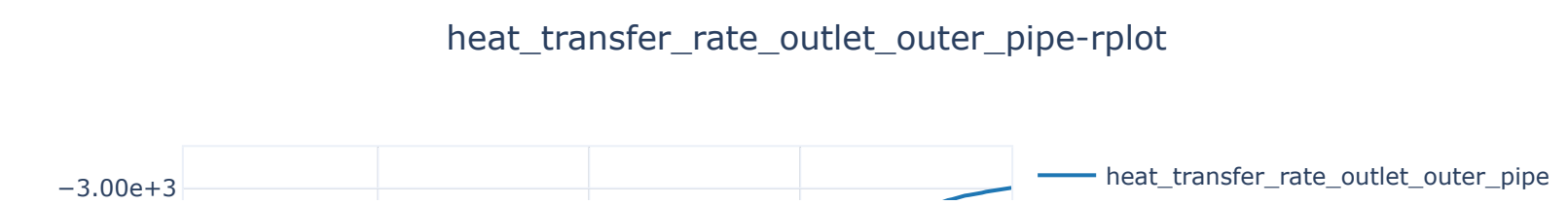
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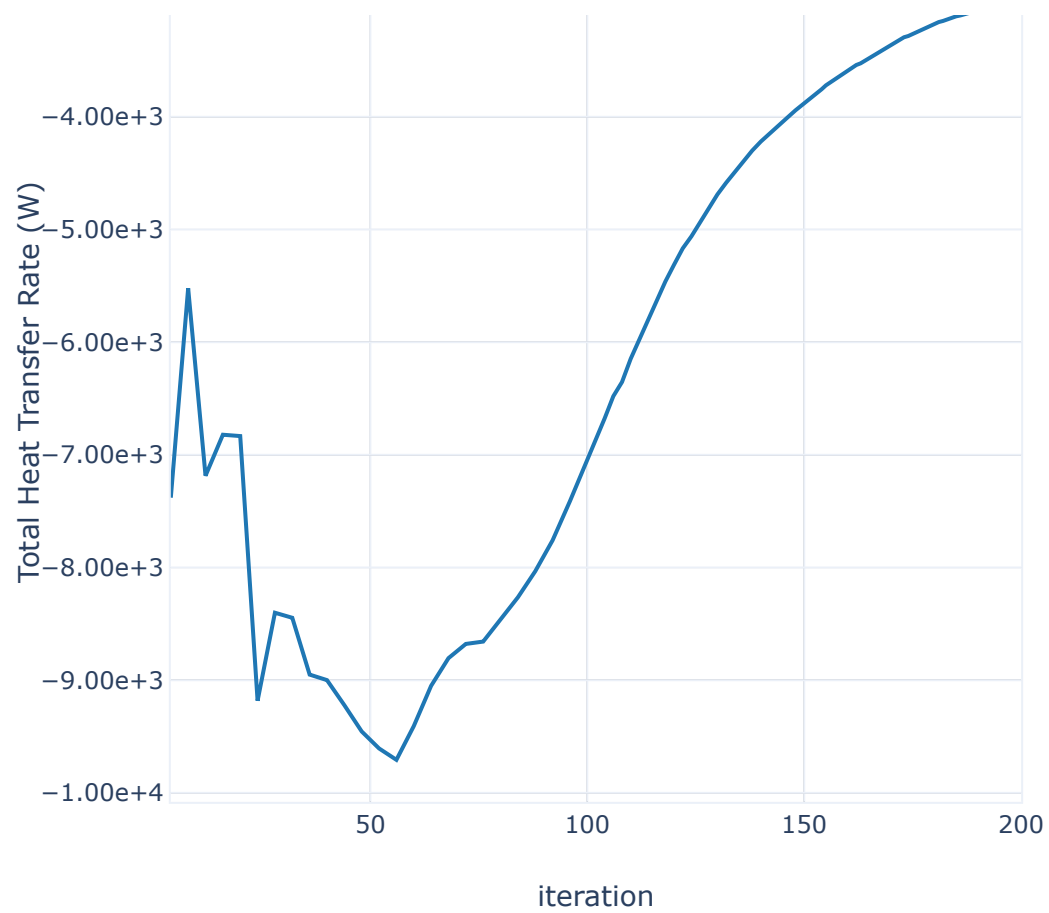


heat_transfer_rate_outlet_inner_pipe-rplot



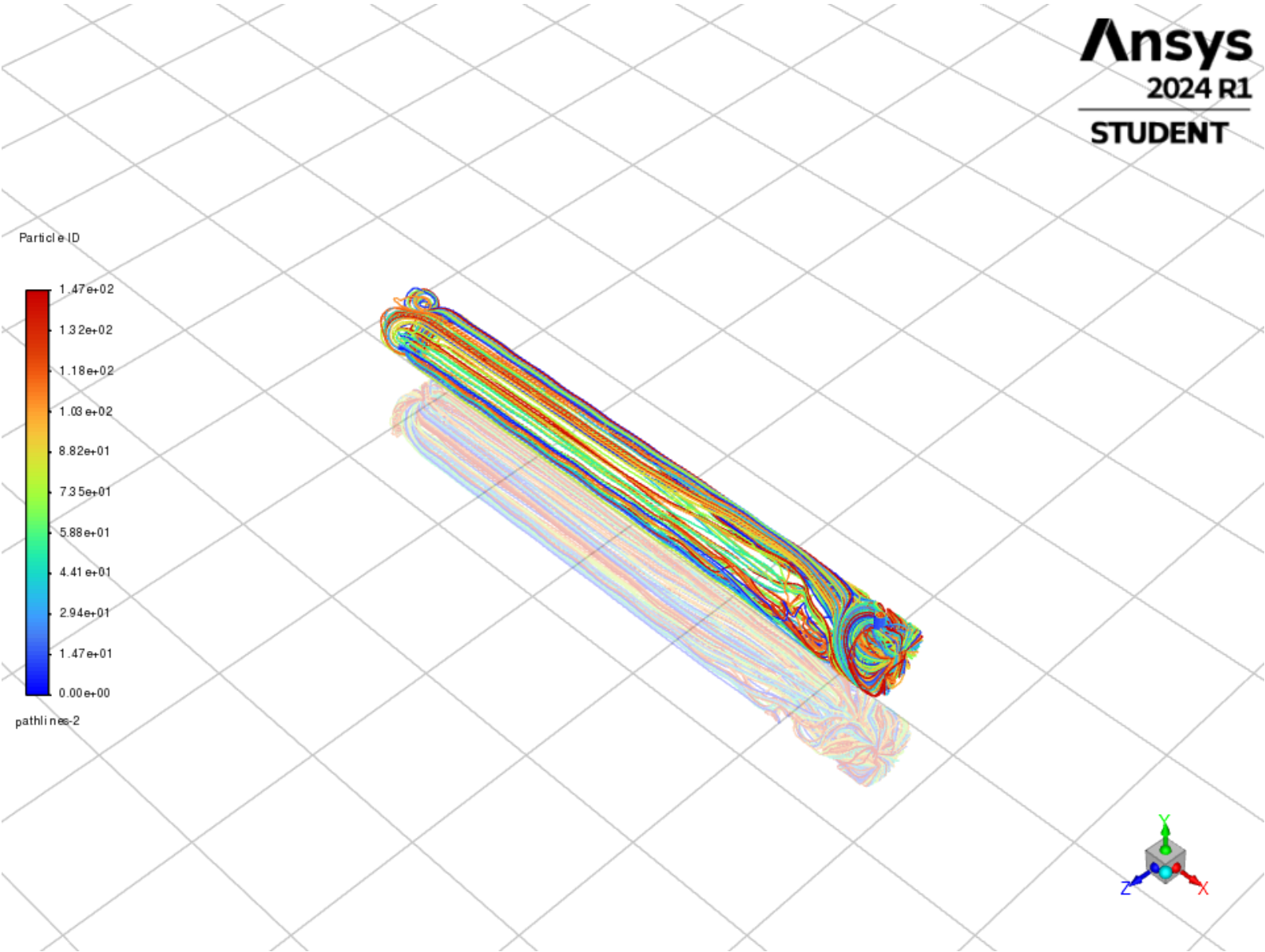
heat_transfer_rate_outlet_outer_pipe-rplot





Pathlines

pathlines-2



pathlines-1

